



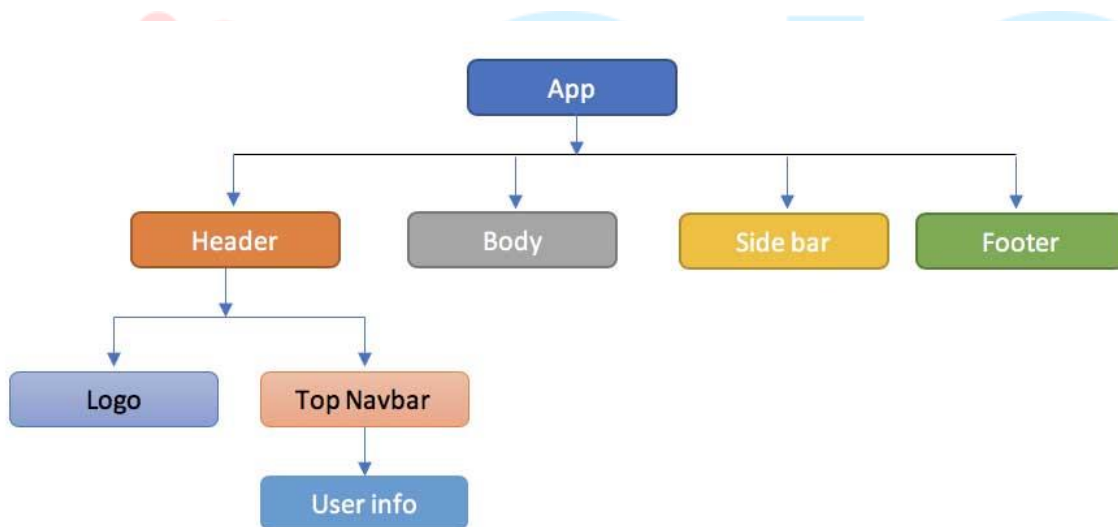
React Components

Component: -

Components are the reusable building blocks of single page application. Components divide our application into several small blocks.

Components forms a tree structure to define or represent our application.

An individual component can re-present a specific part of hole user-interface.



The good react developer must have a skill to divide the hole application into several small parts/pieces, we call them components.

In react there are two types of components.

1. Functional components.
2. Class components.



Functional component: -

React functional components are the simple JavaScript functions, whose name should start with capital letter and which must return some UI (JSX),

React functional component can have props.

Example:

```
function Home()
{
    return (
        <div>
        <h1>React Functional Component</h1>
        <p>lorem*100</p>
        </div>
    );
}
```

export default Home; //to access Home function out-side the

Home.js

Class Components: -

React class components are the JavaScript Classes, which extends the `Component` class of react library. And overrides the `render` function of Component class to return UI (JSX).

The class component can have state.



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Complete Java Classes

by Kunal Sir

Now days, class components are not preferred as compare to functional components, because of performance issue.

Example:

```
import { Component } from "react";

class Demo extends Component
{
    constructor()
    {
        super();
    }
    render()
    {
        return (
            <div>
                <h1>React Class Components...!</h1>
            </div>
        );
    }
}

export default Demo; // to assess Demo class out-side the Demo.js
```



by Kunal Sir

Props: - (properties)

Props are the optional inputs (parameter argument) that are given to the functional component in the form of attributes, while rendering the component as JSX element.

Functional component must have only one parameter argument.

Props can be accessed using first parameter argument of component function. And the first parameter argument of component function always an object.

Except string values, any other value must be pass using { }.

Ex. `<Profile name="pratik" id={101} />`

Example:

1. Course.js: -

```
JS Course.js U X
src > components > course > JS Course.js > [🔗] default
1  import './Course.css';
2
3  function Course(props)
4  {
5      return <div style={{width:"300px",height:'200px',
6          backgroundColor:props.color
7      }}>
8          <h1>Course name: {props.courseName}</h1>
9          <p>Course Fees: {props.fees}</p>
10     </div>
11 }
12
13 export default Course;
```



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2. Home.js: -

```
JS Home.js U x
src > components > JS Home.js > ...
1
2 import Course from "../course/Course.js";
3 function Home()
4 {
5   return <div style={{backgroundColor:'aqua'}}>
6     <h1>I am React Functional Component</h1>
7     <p>some information...!</p>
8     <Course courseName="Java Fullday" fees={95000.00} color="orange"/>
9     <Course courseName="Python Fullday" fees={90000.00} color="royalblue"/>
10
11   </div>
12 }
13
14 export default Home;
```

Task: - Create Below user interface using react functional components.

